

Linear Algebra I

1. Complex roots 2. Diagonalization

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1 Recap

- Eigenvalues and eigenvectors of a matrix
- Complex numbers

2 Complex n th roots

- A motivating example
- Finding the fifth roots of 32

3 Diagonalization

- A 2x2 example

Recap

Eigenvalues and eigenvectors of A

$$A\mathbf{x} = \lambda\mathbf{x} \text{ (and } \mathbf{x} \neq \mathbf{0}\text{)}$$

$$\iff$$

$$(A - \lambda I)\mathbf{x} = \mathbf{0} \text{ (and } \mathbf{x} \neq \mathbf{0}\text{)}$$

Which also implies

- λ is a solution of $\det(A - \lambda I) = 0$
- $\mathbf{x}$ is in $\text{Nul}(A - \lambda I)$

I will often abbreviate the words eigenvalues and eigenvectors to “evalues” and “evectors”

What we've seen so far

Complex numbers

- Some matrices have no real eigenvalues
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$
 - Addition, subtraction, multiplication, division
 - Complex conjugates $\overline{a + bi} = a - bi$
 - Absolute value aka modulus (=length) $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$
 - Argument (=angle from real axis) $\arg(z) = \arctan \frac{b}{a}$
- Every complex polynomial has a solution in $\mathbb{C}$
 - Even the real ones
 - Every matrix (real or complex) has an eigenvalue
 - E.g. $\lambda^2 + 1 = (\lambda + i)(\lambda - i)$ has solutions $\pm i$
 - Fact: Non-real roots of real polynomials appear as conjugate pairs

Learning outcomes

In today's class, we will:

- Investigate n^{th} roots of a complex number, motivated by a question about eigenvalues (Section 5.5)
(Using a 5x5 example matrix)
- Describe the process of diagonalizing a matrix (Section 5.3)
(Using a 2x2 example with real eigenvalues)

Key takeaways today

The #1 takeaways from today's class

- Complex numbers have n distinct n^{th} roots!
(two square roots, three cube roots, four 4th roots, etc.)
(we will see how to find them all)
- If $P = [\text{e vectors of } A]$ and $D = \text{diag}(\text{e values of } A)$, then

$$AP = PD$$

(when P is invertible: $AP = PD \iff A = PDP^{-1}$)

(we say P diagonalizes A) (we call A diagonalizable)

Complex n th roots

Example question

Find all of the (complex) eigenvalues of the matrix A given below.

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

We will compute the characteristic polynomial $\det(A - \lambda I)$ and solve the equation $\det(A - \lambda I) = 0$

Step 1: Compute the characterisitc polynomial:

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \text{ and so } A - \lambda I =$$

Step 2: Solve the characteristic equation

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 32 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \text{ and the characteristic equation of } A \text{ is } \lambda^5 = 32.$$

Question: How many solutions are there to the equation $\lambda^5 = 32$?

Let c be a complex number. To solve

$$z^5 = c$$

- ① Write z and c in polar (exponential) form

$$z = re^{i\theta}, \quad c = se^{i(\tau+2k\pi)}$$

(s and τ are given, we need to find the values r and θ)

- ② Now compute z^5

$$z^5 = (re^{i\theta})^5 = r^5(e^{i\theta})^5 = r^5e^{i(5\theta)}$$

- ③ Equate lengths and angles:

$$r^5 = s \quad \text{and so} \quad r = \sqrt[5]{s} \quad (\text{the real positive 5th root of } s)$$

$$5\theta = \tau + 2k\pi \quad \text{and so} \quad \theta = \frac{\tau}{5} + \frac{2k\pi}{5}$$

The same procedure works for $z^n = w$

Back to the example

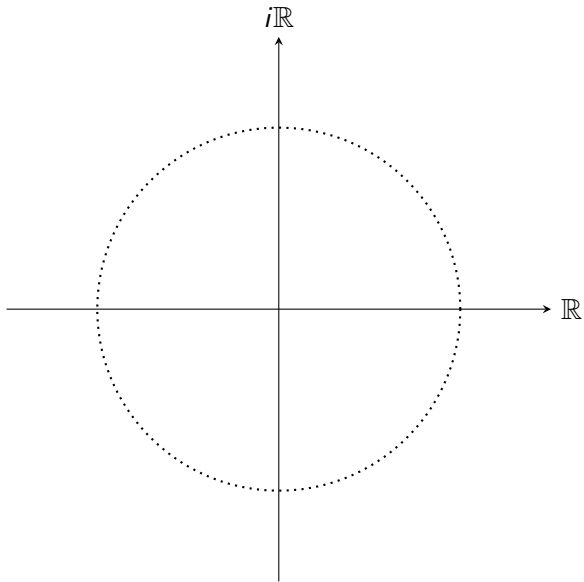
$$\lambda^5 = 32$$

$$\textcircled{1} \quad \lambda = re^{i\theta}, \quad 32 = se^{i(\tau+2k\pi)} =$$

$$\textcircled{2} \quad \lambda^5 = r^5 e^{i(5\theta)}$$

$\textcircled{3}$ Now equate lengths and arguments

Let's draw the solutions:



In a nutshell

If

$$z^n = se^{i\tau}$$

then

- The length of z is the real positive n th root of s
- The argument of z is $\frac{\tau}{5} \pm \frac{2k\pi}{5}$

(This gives five distinct complex roots— they all lie equally spaced around a circle of radius $\sqrt[n]{s}$)

Diagonalization

Find the evectors and values of A

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

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Solution:

$$|A - \lambda I| = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$$

Eigenvalues are 1 and 3

$$\text{Nul}(A - 1I) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

$$\text{Nul}(A - 3I) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

A matrix which diagonalizes A

Diagonalization in general

A (Very) Short History of Complex Numbers

The following timeline of imaginary numbers is for your interest only (non-examinable).

- **Cardano (1500s):** Encountered $\sqrt{-1}$ while solving cubics; used it reluctantly
- **Bombelli (1500s):** First coherent rules for $a + bi$ arithmetic
- **Descartes (1600s):** Coined the term “imaginary numbers” (dismissively)
- **Euler & de Moivre (1700s):** Made complex numbers central in analysis and trigonometry; $e^{i\theta} = \cos \theta + i \sin \theta$
- **Gauss (1800s):** Gave the modern viewpoint — complex plane, geometric meaning, and rigorous use

Complex numbers weren't invented all at once — they gradually became indispensable